

DATA VORTEX A'26

Round 2 — Technical Report

Rebuilding the Social Engine — Semantic Understanding Layer

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1. Executive Summary

Round 2 rebuilds the Social Engine's semantic understanding layer using Natural Language Processing. Dataset 2 contains 9,000 labelled social-media text records with supervised sentiment and topic targets.

The pipeline uses grouped stratified splitting, text normalization, word and character TF-IDF, classical linear models, imbalance handling, hyperparameter experiments, and DistilBERT fine-tuning. The final solution is task-specific: DistilBERT for sentiment and combined word/character TF-IDF with balanced Logistic Regression for topic classification.

Final held-out performance: sentiment accuracy 70.06%, Macro F1 0.6988; topic accuracy 94.22%, Macro F1 0.6939.

2. Problem Definition

The objective is to classify each social-media post by sentiment (Negative, Neutral, Positive) and topic (Account_Security, Community_Discussion, Feature_Feedback, Technical_Issues). Topic classification is strongly imbalanced, so Macro F1 is emphasized.

3. Dataset and EDA

The notebook records 9,000 rows and four columns. There are no missing values or fully duplicated rows. There are 7,900 unique post texts and 1,100 repeated-text records. No displayed duplicate texts had conflicting labels.

Text length ranges from 4 to 35 words, with a mean of approximately 19.47 words.

Target	Class	Count	Share
Sentiment	Negative	3,000	33.33%
Sentiment	Neutral	3,000	33.33%
Sentiment	Positive	3,000	33.33%
Topic	Community_Discussion	7,752	86.13%
Topic	Technical_Issues	815	9.06%
Topic	Feature_Feedback	297	3.30%
Topic	Account_Security	136	1.51%

4. Data Split and Leakage Control

StratifiedGroupKFold with five folds, shuffle=True and random_state=42 was used. The joint sentiment/topic label was used for stratification and post_text was used as the grouping variable. Training contains 7,183 rows and testing contains 1,817 rows; text overlap is zero.

Partition	Rows	Share	Unique texts
Training	7,183	79.81%	6,320
Testing	1,817	20.19%	1,580
Text overlap	0	—	0

5. Preprocessing Pipeline

Text normalization lowercases text, removes URLs, replaces @mentions with USERMENTION, collapses whitespace, and trims text. Stopwords were not aggressively removed.

Word TF-IDF used unigrams/bigrams with min_df=2, max_df=0.95 and sublinear TF. Character TF-IDF used 3–5 character n-grams with min_df=2, max_features=30,000 and sublinear TF. Combined dimensionality was 55,641 features.

DistilBERT tokenization used distilbert-base-uncased with maximum sequence length 64.

6. Model Selection

Task	Model	Accuracy	Macro F1
Sentiment	Logistic Regression	59.71%	0.5971
Sentiment	Linear SVM	59.27%	0.5919
Topic	Logistic Regression	87.95%	0.3414
Topic	Linear SVM	91.41%	0.5465

Task	Model	Accuracy	Macro F1
Sentiment	Balanced Logistic Regression	59.99%	0.5993
Sentiment	Balanced Linear SVM	59.16%	0.5908
Topic	Balanced Logistic Regression	91.36%	0.5985
Topic	Balanced Linear SVM	91.91%	0.5783

Experiment	Task	Accuracy	Macro F1	Decision
Combined Word + Char TF-IDF + Balanced LR	Sentiment	61.14%	0.6121	Used as classical baseline
Combined Word + Char TF-IDF + Balanced SVM	Sentiment	60.43%	0.6050	Rejected
Custom class-weighted LR	Topic	91.74%	0.5225	Rejected
Word trigrams + Char TF-IDF	Sentiment	60.98%	0.6107	Rejected
Word trigrams + Char TF-IDF	Topic	94.11%	0.6804	Rejected
Tuned LR, C=2	Sentiment	61.36%	0.6147	Retained before transformer stage
Tuned LR, C=0.25	Topic	93.62%	0.6797	Rejected on held-out Macro F1
ComplementNB	Sentiment	61.59%	0.6098	Rejected
ComplementNB	Topic	86.46%	0.2866	Rejected

7. Transformer Modelling

The notebook records GPU training on a Tesla T4 using PyTorch 2.11.0+cu128, Transformers 5.16.1, Datasets 4.8.5 and Accelerate 1.14.0. DistilBERT was trained separately for sentiment and topic classification using three epochs, learning rate 2e-5, weight decay 0.01, train batch size 16, evaluation batch size 32, FP16, and Macro F1-based checkpoint selection.

DistilBERT improved sentiment performance. For topic, it achieved high accuracy but weaker Macro F1 because Feature_Feedback was predominantly classified as Community_Discussion; therefore the classical topic model was retained.

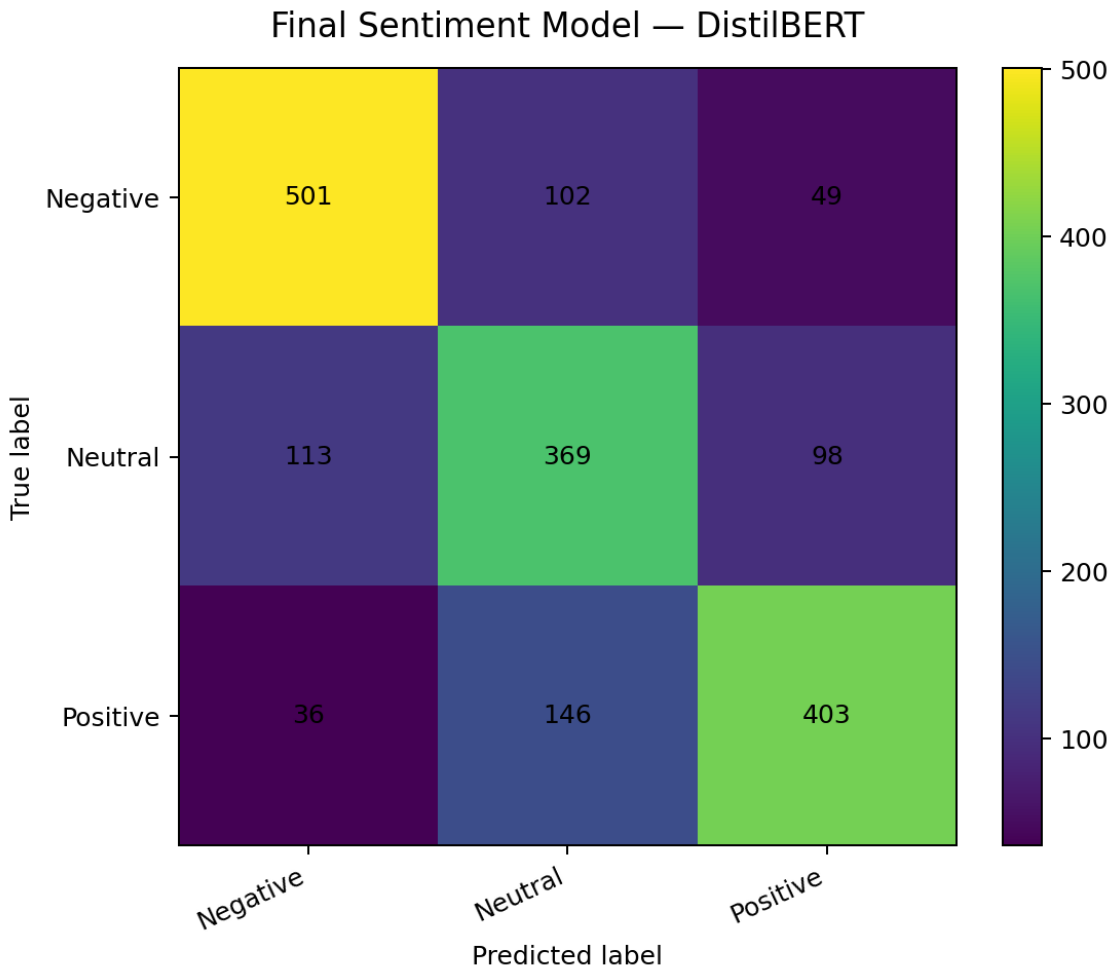
8. Final Model Selection

Task	Final model	Accuracy	Macro F1
Sentiment	DistilBERT (distilbert-base-uncased)	70.06%	0.6988
Topic	Word + Character TF-IDF +	94.22%	0.6939

9. Evaluation and Confusion Matrices

Sentiment classification report:

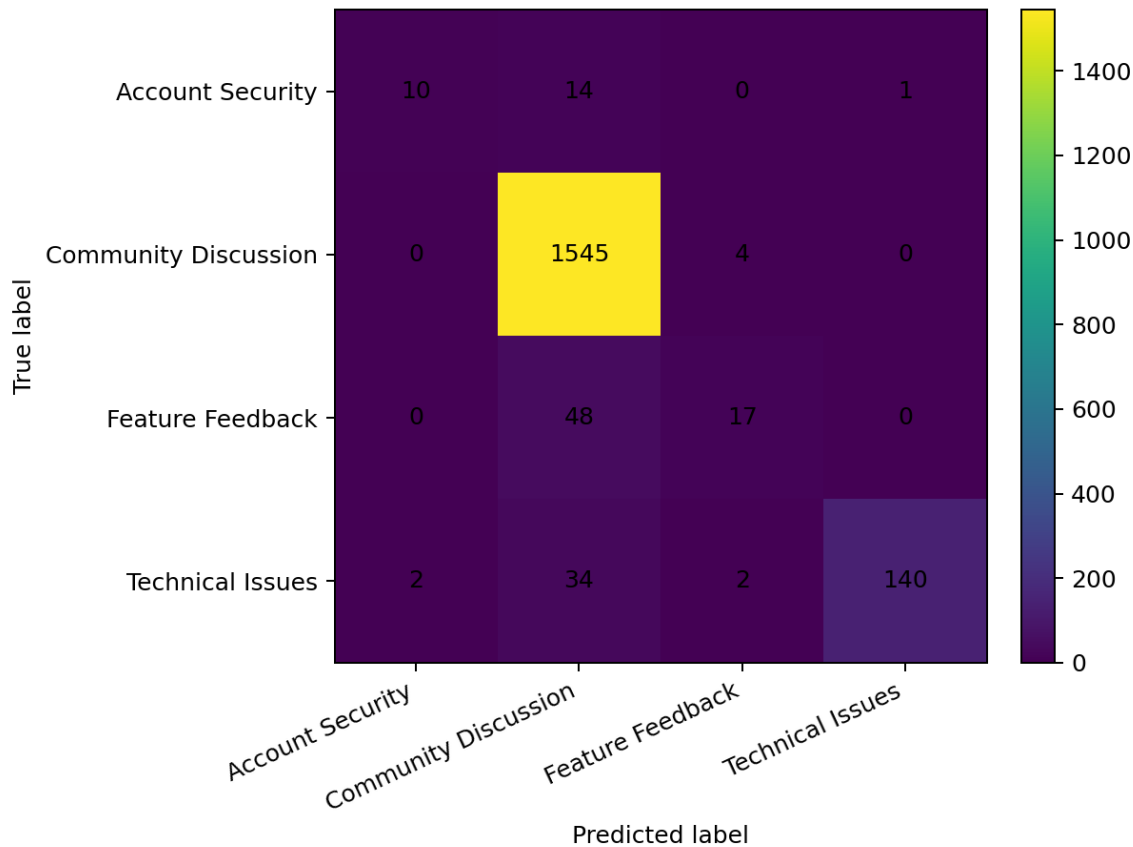
Class	Precision	Recall	F1	Support
Negative	0.7708	0.7684	0.7696	652
Neutral	0.5981	0.6362	0.6165	580
Positive	0.7327	0.6889	0.7101	585
Macro Avg	0.7005	0.6978	0.6988	1817
Weighted Avg	0.7034	0.7006	0.7016	1817



Topic classification report:

Class	Precision	Recall	F1	Support
Account_Security	0.83	0.40	0.54	25
Community_Discussion	0.94	1.00	0.97	1549
Feature_Feedback	0.74	0.26	0.39	65
Technical_Issues	1.00	0.79	0.88	178

Final Topic Model — TF-IDF + Balanced Logistic Regression



10. Error Analysis

Sentiment errors: 544. The most common transitions are Positive→Neutral (146), Neutral→Negative (113), Negative→Neutral (102), Neutral→Positive (98), Negative→Positive (49), and Positive→Negative (36).

Topic errors: 105. The dominant failures are Feature_Feedback→Community_Discussion (48), Technical_Issues→Community_Discussion (34), and Account_Security→Community_Discussion (15).

11. Processing Pipeline Design

Dataset → validation/EDA → grouped stratified split → text normalization → TF-IDF features → classical model comparison → imbalance handling → feature/hyperparameter experiments → transformer fine-tuning → held-out evaluation → confusion matrices → error analysis → model serialization.

12. Limitations and Future Improvements

The main limitation is severe topic imbalance. Future work could use targeted minority-class augmentation, domain-adapted pretrained models, threshold calibration, hierarchical classification, and additional minority

labels. Sentiment could benefit from domain adaptation and improved handling of negation, slang, emojis and contextual references.

13. Conclusion

Round 2 produced a reproducible NLP semantic classification pipeline and selected models separately for sentiment and topic based on held-out evaluation and error analysis.